

Biology All In One (BAIO) Learning System

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Chapter 9: Biotechnology – Principles and Processes

Worksheet · Bioreactors & Downstream Processing

I. Multiple Choice Questions

- Introduction of recombinant DNA into bacterial cells is typically achieved by:
(A) Transformation (B) Photosynthesis induction
(C) Meiosis (D) Conjugation only
- Introduction of recombinant DNA into plant and animal cells is achieved using:
(A) Transformation only (B) Disarmed vectors, biolistics, and micro-injection
(C) Only antibiotic selection (D) PCR amplification
- Large vessels used for large-scale culture of recombinant cells are called:
(A) Petri dishes (B) Bioreactors
(C) Test tubes (D) Gel tanks
- Typical industrial bioreactors hold culture volumes of about:
(A) 1-5 litres (B) 100-1000 litres
(C) 1 million litres always (D) A few millilitres only
- Which culturing method keeps cells continuously in their log/exponential phase for higher yield?
(A) Batch culture (B) Continuous culture
(C) Static culture (D) Anaerobic culture

II. Fill in the Blanks

- Recombinant DNA is introduced into bacteria mainly by _____.
- Plant and animal cells receive recombinant DNA via disarmed vectors, biolistics, and _____.
- Large vessels used for large-scale culture production are called _____.
- Industrial bioreactors typically hold culture volumes of about _____ litres.
- In _____ culture, used media is drained and fresh media is added to keep cells in log phase.
- Separation, purification, and finishing of a crude bioreactor product is called _____ processing.

III. Match the Following

Column A	Column B
1. Agitator	a. Withdraws culture for monitoring
2. Sparger	b. Bubbles air into the reactor
3. Foam control system	c. Mixes contents, distributes oxygen
4. Sampling port	d. Prevents excess foaming

IV. True / False

- Bioreactors typically hold only 1–5 litres of culture. [True / False]

2. Continuous culture keeps cells in their log/exponential growth phase. [True / False]
3. Downstream processing occurs inside the bioreactor before the crude product forms. [True / False]
4. Finished bioproducts must undergo clinical trials and quality checks before reaching market. [True / False]
5. A foam control system is a standard component of a bioreactor. [True / False]

V. 2 Marks Questions

1. List the main components of a bioreactor and briefly describe each one's function.

2. Explain how continuous culture helps achieve a higher product yield.

VI. 3 Marks Questions

1. What is downstream processing, and why is it necessary?

2. What must a finished bioproduct undergo before it can be sent to market?

VII. 5 Marks Questions

1. With the help of a labelled diagram, describe the structure and function of a bioreactor, and explain the significance of continuous culture in large-scale production.
